Status Report N

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Status Update

This status update focuses on the first-round optimization results for d0-d1 parameter tuning. We have completed a reproducible sweep pipeline under fixed placement and a unified analysis workflow. The current goal is to use link-level metrics to select representative parameter sets, then perform spectrum-based cross-validation in the lab.

Input from previous discussion.

From the previous discussion, we agreed on three points: (1) use single-variable sweeps under fixed test conditions; (2) evaluate changes primarily with BER and RSSI while excluding empty and low-sample runs; (3) validate baseline, clearly improved, and clearly degraded parameter sets with frequency-domain evidence in the lab.

Current implementation status.

We have completed the first round of system-level scanning over d0-d1 parameter combinations under a fixed physical setup. Our implementation strategy is to tune backscatter frequency separation and center-offset-related settings to improve receiver distinguishability and link stability. For receiver bandwidth, we follow the rule of selecting a preset that is not below the minimum required bandwidth, and choosing the next higher preset level to reduce clipping-related misinterpretation. At this stage, BER and RSSI are used as the main impact metrics. We also added parameter-space visualization (BER/RSSI heatmaps) to capture global trends rather than relying on isolated points.

Preliminary results.

In the current dataset, we have completed unified processing for valid parameter combinations and removed empty files and low-sample runs to avoid low-quality conclusions. Based on BER and RSSI parameter-distribution plots, we identified three representative classes: baseline (20-18), clearly improved groups (e.g., 24-22 / 36-32), and clearly degraded groups (e.g., 32-26). Preliminary observations show that parameter tuning significantly affects link behavior, but RSSI and BER are not always aligned. Therefore, single-metric interpretation is insufficient. The current conclusion is that parameter optimization is effective, but frequency-domain evidence is still required to explain why specific settings improve or degrade performance.

Next steps.

Next, we will collect spectrum captures in the lab for baseline, improved, and degraded parameter sets. We will focus on center offset, sideband separation (deviation), occupied bandwidth, and sideband-to-carrier level. These observations will be cross-checked against BER/RSSI trends under the same setup. If frequency-domain and link-level results are consistent, the corresponding parameter set will be promoted as a default candidate for implementation.

Experimental optimisation

Objective and hypothesis

This optimization stage focuses on d0-d1 parameter combinations. Since d0-d1 determines sideband spacing and center-offset-related behavior, it directly affects symbol distinguishability at the receiver. Our hypothesis is that there exists an intermediate parameter region where BER improves clearly over baseline while RSSI remains stable; both overly narrow and overly wide spacing may reduce robustness.

Experimental method

We use a fixed-setup, single-variable sweep with a unified analysis pipeline. Specifically, we keep placement, antenna orientation, baud rate, and packet flow unchanged; sweep d0-d1 combinations; parse logs with one script; compute BER and RSSI consistently; and exclude empty/low-sample runs before comparison. Results are visualized as parameter-space heatmaps to avoid overfitting decisions to single points.

Results and description

The first-round sweep shows structured performance variation across parameter combinations. BER and RSSI do not always move in the same direction, which confirms that multi-metric interpretation is necessary. At this stage, we have a practical shortlist of representative sets for further validation: one baseline set, improved candidates, and degraded references. This narrows the design space for controlled lab validation.

Table 1: Representative parameter-set comparison (baseline / improved / degraded).

Parameter set (d0-d1)	BER (%)	RSSI (dBm)	Deviation (kHz)	Note
20-18	17.71	-98.60	347.22	Baseline
24-22	7.87	-94.97	236.74	Improved candidate
36-32	6.67	-96.87	217.01	Improved candidate
32-26	19.40	-99.59	450.72	Degraded reference

Discussion and stage conclusion

The current results indicate that d0-d1 tuning produces measurable and reproducible link-level differences. At this stage, 24-22 and 36-32 are prioritized as candidate settings for lab validation. Final parameter selection will be based on consistency between frequency-domain observations and BER/RSSI outcomes, not on a single metric alone.

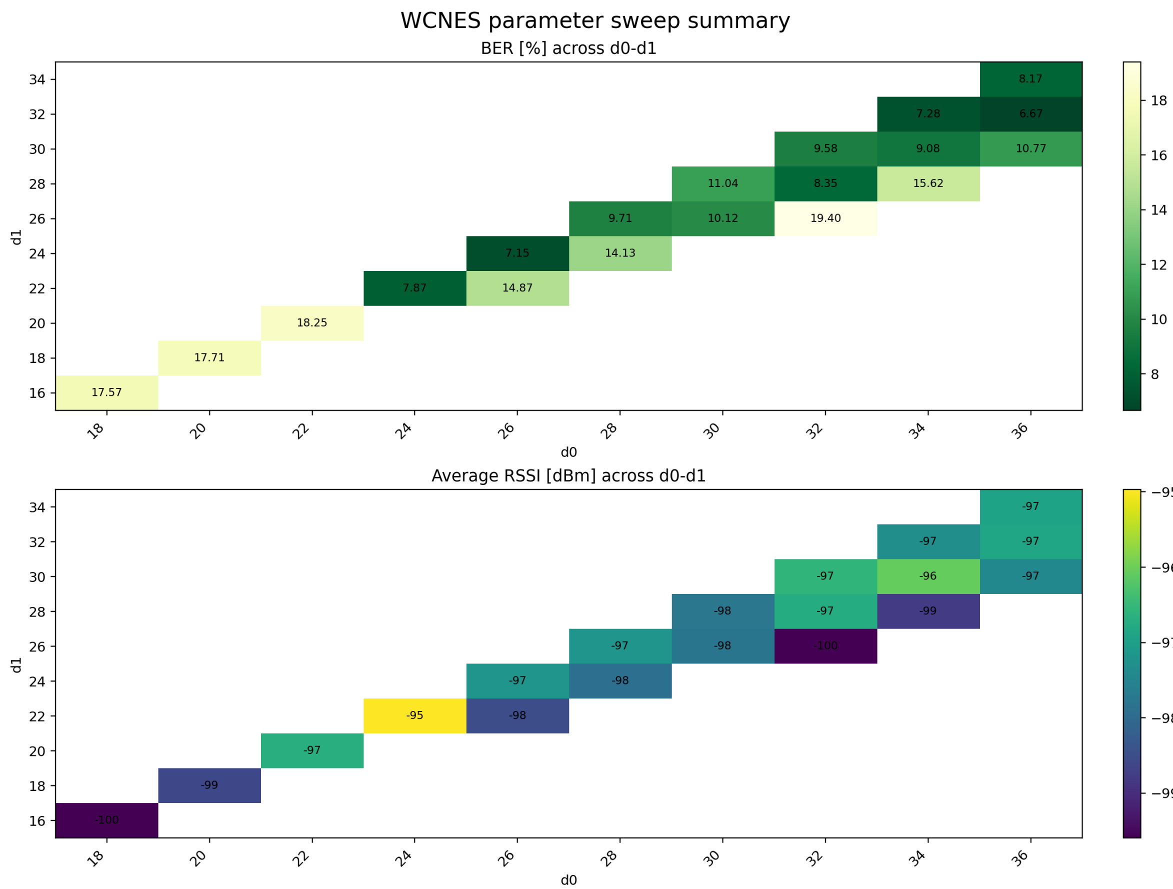

Figure 1: BER and RSSI distributions over d0-d1 parameter combinations.

References